

Determining how the pH of Coca-Cola Original affects its caffeine content

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INTRODUCTION

Coca-Cola Original is a carbonated soft drink produced by dissolving carbon dioxide into water and adding flavorings, sweeteners, and caffeine, resulting in a highly acidic beverage (*The Coca-Cola Company*). Coca-Cola is my favourite comfort drink that I consume when I need a quick boost of energy after long school days. Coca-Cola has a low pH of approximately 2.5 due to the presence of phosphoric acid and carbonic acid, making it significantly more acidic than many other common beverages such as tea or coffee. Since caffeine is a weak base, I became curious about how changes in acidity, by altering the pH, might influence its caffeine concentration.

RESEARCH QUESTION

This exploration aims to establish: How does the pH of Coca-Cola Original affect its caffeine content?

THEORETICAL RELATIONSHIP AND HYPOTHESIS

Caffeine is an alkaloid with the chemical formula $C_8H_{10}N_4O_2$. It is a weak base due to the presence of nitrogen atoms in its structure, meaning its solubility and stability can be influenced by the pH of its surroundings. Changes in pH may affect the protonation state of caffeine, which could influence its apparent concentration when analyzed. As the pH of Coca-Cola increases, caffeine may become less protonated, potentially affecting its solubility or interaction with the solvent. This chemical behavior leads me to the hypothesis that increasing the pH of Coca-Cola Original may result in a decrease in the amount of measured caffeine concentration.

Caffeine

Caffeine ($C_8H_{10}N_4O_2$) is one of the most widely consumed psychoactive stimulants, commonly ingested through beverages such as coffee, tea, and carbonated soft drinks like Coca-Cola. Caffeine is an organic alkaloid with a molecular weight of 194.19 g/mol. In pure form, it is a bitter white powder. Although caffeine occurs naturally in coffee beans and tea leaves, in soft drinks it is added during manufacturing. When consumed, the pharmacological effects of caffeine act to stimulate the central nervous system, sustaining intellectual activity, and decreasing reaction times.

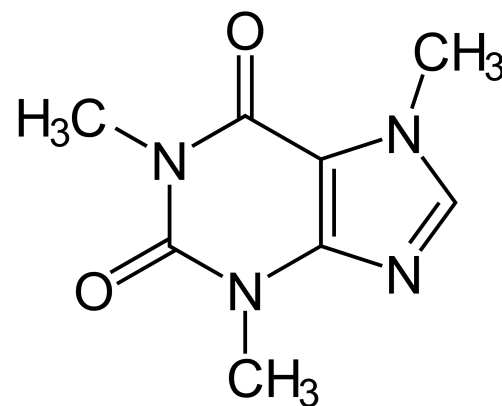


Figure 1. Molecular structure of caffeine

Structurally, caffeine (1,3,7-trimethylxanthine) is an alkaloid containing three methyl groups attached to nitrogen atoms at positions 1, 3, and 7, as shown in Figure 1. The presence of two carbonyl ($C=O$) functional groups contributes to its polarity. Due to the nitrogen atoms within its heterocyclic structure, caffeine behaves as a weak base, meaning its protonation state can change depending on the pH of the

solution. This property is particularly relevant in acidic beverages such as Coca-Cola, as variations in pH may influence caffeine's solubility and behavior during extraction.

Ethyl Acetate

Ethyl acetate ($\text{CH}_3\text{COOC}_2\text{H}_5$) is a volatile organic compound and one of the simplest carboxylate esters, formed through the esterification of ethanol and acetic acid. It is a colorless, flammable liquid, and an explosion hazard. It is slightly soluble in water, but soluble in most organic solvents. Ethyl acetate is commonly used as a solvent in paints, perfumes, and cleaning products (*Solventis Ltd*). Ethyl acetate is also used in the decaffeination process of beverages because of its ability to dissolve caffeine effectively. This is reflected in the molecular structure of ethyl acetate (Figure 2) which contains both polar and nonpolar regions, allowing it to interact with moderately polar compounds such as caffeine.

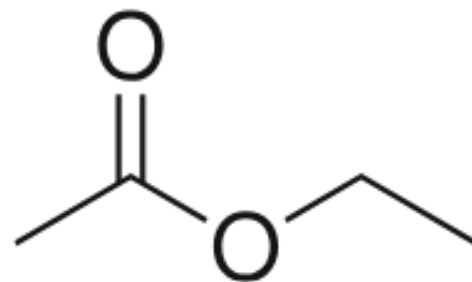


Figure 2. Molecular structure of ethyl acetate

Extraction Method

In this investigation, liquid-liquid extraction using ethyl acetate was employed to isolate the caffeine from Coca-Cola Original. Ethyl acetate was selected because caffeine is more soluble in organic solvents than in water, particularly when it is in its neutral (less protonated) form. Since caffeine is a weak base, altering the pH of the Coca-Cola may influence the extent to which caffeine partitions into the ethyl acetate layer (*Westfield State University, n.d*).

Ethyl acetate is immiscible with water, meaning the two liquids do not mix, and instead form two distinct layers. Because ethyl acetate has a lower density than water, it forms the top layer, while the aqueous Coca-Cola solution remains as the bottom layer. During extraction, caffeine preferentially dissolves into the ethyl acetate layer due to favorable intermolecular interactions (*Libretexts, n.d.*). A separatory funnel is used to separate these two layers, collecting the top layer with the desired caffeine. Figure 3, labelled, illustrates the formation of the two immiscible layers in the separatory funnel.

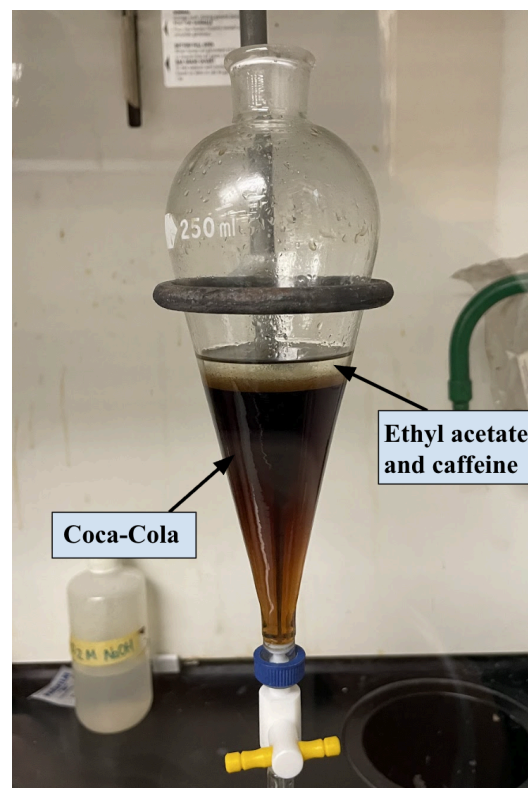


Figure 3. Separatory Funnel

VARIABLES

Variable Type	Variable	Description
Independent	pH	The pH of the Coca-Cola is measured with a calibrated pH meter with the use of Vernier LabQuest. The pH meter was submerged into the Coca-Cola until the pH stabilized at a reading. Measured amounts of 2.0 mol dm^{-3} sodium hydroxide or 6.0 mol dm^{-3} hydrochloric acid were added dropwise to separate Coca-Cola samples.
Dependent	Caffeine content (g)	Caffeine was extracted from the Coca-Cola with ethyl acetate and evaporated in a beaker until all the solution was gone. The difference of the mass of the beakers determines the mass of the caffeine in each trial.
Controlled	Sitting and degassing time	Before the pH adjustment and measuring, each Coca-Cola sample was left to sit uncovered for a fixed period of time (approximately 24 hours) to allow dissolved carbon dioxide to escape (degassing) before caffeine extraction. It was also lightly stirred. This was done as carbon dioxide present in carbonated beverages can interfere with accurate volume measurements and pH stability.
	Brand and type of soda	The same type and brand of Coca-Cola was used for all samples: Coca-Cola Original Taste (aka Coca-Cola Classic). This was done because different formulations may contain different initial caffeine concentrations. Using the same product ensures a consistent starting amount of caffeine.
	Volume of Coca-Cola in each trial (cm^3)	200 cm^3 of distilled Coca-Cola was measured to conduct each trial. Varying volumes would change the total amount of caffeine present when taking an aliquot of the sample.
	Temperature of the samples	All Coca-Cola samples were kept at room temperature throughout the experiment. Temperature can affect solubility and extraction efficiency, which could alter the measured caffeine content.
	Time allowed between pH adjustment and measurement	Caffeine extraction was performed immediately after pH adjustment for every sample as allowing different waiting times could lead to evaporation or chemical changes that may affect caffeine concentration.
	Volume of ethyl acetate (cm^3)	Three iterations of 10 cm^3 ethyl acetate were measured and mixed into the separatory funnel to ensure that all the caffeine was extracted from each sample with an equal comparison.

LAB EQUIPMENT

1. Erlenmeyer flasks
 - a. six 250 cm^3 ($\pm 0.01\text{cm}^3$)
2. beakers
 - a. one 600 cm^3
 - b. three 150 cm^3
 - c. six 50 cm^3
3. 100 cm^3 graduated cylinder
4. 250 cm^3 separatory funnel
5. Wash bottle with distilled water
6. Ring clamp
7. Ring stand
8. Vernier LabQuest
9. pH meter
10. one eye dropper
11. Six 330 mL Coca-Cola Original soda cans
12. Distilled water

CHEMICALS

13. 200 cm^3 of ethyl acetate
14. 30 cm^3 of 6.0 mol dm^{-3} hydrochloric acid
15. 10 cm^3 of 1.0 mol dm^{-3} sodium hydroxide

PROCEDURE

1. 200 cm^3 of Coca-Cola Original was poured into a 250 cm^3 beaker.
2. Degassing at room temperature (approximately 22.0°C) through stirring and leaving to sit for 24 hours.
3. A pH meter was placed in the Coca-Cola to measure and record the pH (it was rinsed before and after use with distilled water).
4. The Coca-Cola was transferred into a separatory funnel from the beaker using a funnel.
5. 10 cm^3 of ethyl acetate was measured in a 100 cm^3 graduated cylinder and poured into the separatory funnel.
6. When two distinct layers formed, the separatory funnel was opened to capture the bottom layer of Coca-Cola in a 250 cm^3 erlenmeyer flask.
7. The Coca-Cola was poured back into the closed separatory funnel.
8. Steps 5-7 were repeated two more times.
9. The remaining layer was collected in a 250 cm^3 beaker and left to evaporate until only caffeine was left.
10. A 50 cm^3 beaker was weighed.
11. 10 cm^3 of ethyl acetate was poured into the 250 cm^3 beaker containing the caffeine to collect it, and then poured into the 50 cm^3 beaker.

12. The 50 cm³ beaker was left to evaporate until all the solution was gone (approximately 2 days to ensure it completely evaporated).
13. The resulting mass of the 50 cm³ beaker was recorded.
14. Steps 1–16 were repeated four additional times, but the pH of the Coca-Cola was adjusted by dropping 2.0 mol dm⁻³ hydrochloric acid or 6.0 mol dm⁻³ sodium hydroxide to achieve pH values of 0.99, 4.5, 6.91, 9.18, 11.65.

SAFETY CONCERNS AND WASTE DISPOSAL

All the beakers, sitting solutions caffeine, pH-adjusted Coca-Cola, and the separatory funnel sat in the fume hood at all times. To ensure proper disposal of ethyl acetate, all glassware and equipment that came into contact with ethyl acetate were rinsed with distilled water, and the rinse was collected in a sealed organic waste container located in the fume hood. Only after this, the glassware was washed thoroughly in the lab sink. This procedure minimized the release of hazardous organic solvents into the environment.

The hydrochloric acid and sodium hydroxide solutions used to adjust the pH of the samples were neutralized by mixing them together, allowing a neutralization reaction to occur and forming water and a salt. The resulting solution was diluted with water and safely disposed of down the drain in accordance with laboratory safety guidelines.

Throughout the experiment, safety goggles and gloves were worn at all times to prevent accidental contact with chemicals.

QUANTITATIVE DATA

Table 1. Raw Data

pH (± 0.01)	0.99	2.37	4.5	6.91	9.18	11.65
Initial mass of 50 cm ³ beaker (± 0.001 g)	30.139	29.523	30.620	29.431	29.573	29.632
Final mass of 50 cm ³ beaker (± 0.001 g)	30.120	29.540	30.635	29.442	29.575	29.633

Table 2. Processed Data

pH (± 0.01)	0.99	2.37	4.5	6.91	9.18	11.65
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Caffeine ($\pm 0.002\text{ g}$)	0.019	0.017	0.015	0.011	0.002	0.001
% Uncertainty of Caffeine Mass	10.52%	11.76 %	13.33%	18.18%	100.0%	200.0%

* The percentage uncertainty of caffeine mass was calculated to four significant figures for accuracy and consistency. *

CALCULATIONS

Table 3. Sample Calculations

Mass of caffeine
Sample calculation of the mass of caffeine for the variation with pH 2.37 (original pH): Mass of caffeine = final mass of the beaker - initial mass of the beaker Mass of caffeine = 29.540 - 29.523 Mass of caffeine = 0.017 g
Uncertainty of mass of caffeine
Sample calculation of the uncertainty for the mass of caffeine: Uncertainty = uncertainty of initial mass of the beaker + uncertainty of final mass of the beaker Uncertainty = 0.001 + 0.001 Uncertainty = 0.002 g
Percent uncertainty of mass of caffeine
Sample calculation of % uncertainty of caffeine mass: $\% \text{ Uncertainty} = \frac{\text{uncertainty mass of caffeine}}{\text{mass of caffeine}} \times 100$ $\% \text{ Uncertainty} = \frac{0.002}{0.017} \times 100$ % Uncertainty = 11.76 %

THEORETICAL AND EXPERIMENTAL VALUES

On the Coca-Cola Company website, it states that for a 330 mL can of Coca-Cola Classic, there is 32 mg or 0.032 g of caffeine. This theoretical value allows a comparison to the results of this experiment (the experimental value). The original pH that did not undergo a change was the Coca-Cola with a pH of 2.37. Thus, the percent error that compares the experimental and literature values can be seen below, using pH of 2.37 as an example:

Table 4. Theoretical and Experimental values

Theoretical value	Experimental value
330 mL: 32 mg	250 mL : 17 mg
To ensure a proportional ratio with the experimental values, the volume of the Coca-Cola must first be multiplied by 1.32 in order to have the same volume as the theoretical. Multiplying the experimental value by 1.32 creates a ratio of 330 mL : 22.44 mg. Afterwards, the values can be substituted into the equation to find percentage error.	
Theoretical value	Experimental value
330 mL : 32 mg	330 mL : 22.44 mg
$\% \text{ Error} = \frac{\text{experimental mass} - \text{theoretical mass}}{\text{literature mass}} \times 100$ $\% \text{ Error} = \frac{22.44 - 32}{32} \times 100$ $\% \text{ Error} = - 29.88 \%$	

Thus, the negative value of percentage error shows that the experimental value is lower than the theoretical value, meaning there may have been errors throughout the experiments that impacted the results to be lower than the literature value. Experimental limitations could include incomplete extraction of caffeine, loss of substance during transfers between containers, or incomplete evaporation of ethyl acetate. Since this literature value is only able to be compared to the mass of caffeine obtained from the Coca-Cola that did not undergo a pH change, it is assumed that the other errors are of a similar size.

GRAPHICAL ANALYSIS

Caffeine (g) vs. pH

● Caffeine (g) — Linear Line of Best Fit for Caffeine (g) $R^2 = 0.947$

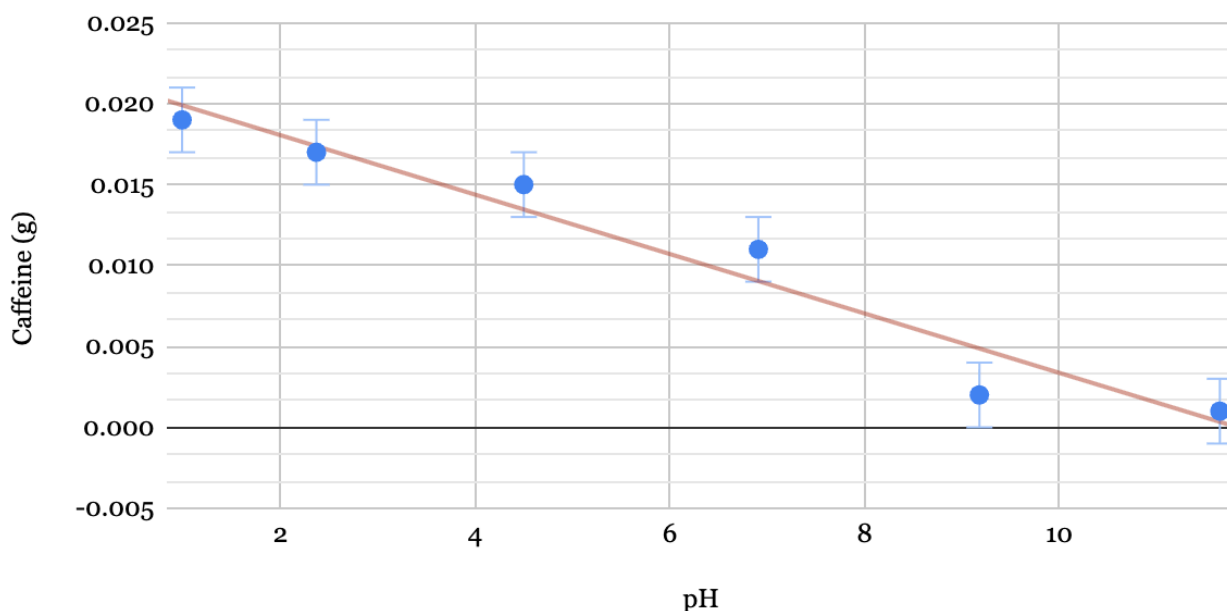


Figure 4. Graph of Caffeine (g) against pH

Errors bars	R ² value
The vertical error bars represent the constant absolute uncertainty in the mass of caffeine extracted (± 0.002 g). This uncertainty results from the measuring instrument precision of ± 0.001 g for both the initial and final mass measurements.	The coefficient of determination (R^2) for the linear regression was 0.947, indicating that approximately 94.7% of the variation in the measured caffeine mass can be explained by changes in pH. This suggests a strong negative linear relationship between pH and the amount of caffeine extracted.

ANALYSIS

Figure 4 displays the relationship between the pH of Coca-Cola and the mass of caffeine extracted and the independent, tested pH values of 0.99, 2.37, 4.50, 6.91, 9.18, and 11.65. The graph shows a clear negative linear correlation between pH and the measured caffeine mass. As the pH of the Coca-Cola increased, the amount of caffeine recovered decreased. This suggests

that more acidic conditions favor the extraction or recovery of caffeine, while increasingly basic conditions result in smaller measured caffeine masses. The most acidic sample (pH 0.99) produced the largest measured mass of caffeine (0.019 g), while the most basic sample (pH 11.65) produced the smallest mass (0.001 g). The line of best fit further supports this relationship, with a coefficient of determination (R^2) of 0.947. This indicates that approximately 94.7% of the variation in caffeine mass can be explained by changes in pH, suggesting a strong negative linear correlation between the two variables. The overall trend in the data suggests that higher pH values lead to lower measured caffeine content, indicating that the pH of Coca-Cola influences the effectiveness of caffeine extraction using ethyl acetate.

CONCLUSION

The aim of this investigation was to determine how the pH of Coca-Cola Original affects the mass of caffeine recovered during extraction. The results from this experiment show a clear negative relationship between pH and the mass of caffeine extracted. This suggests that caffeine extraction using ethyl acetate is more effective under acidic conditions than under basic conditions. Overall, the data suggest that increasing the pH of Coca-Cola reduces the amount of caffeine extracted.

STRENGTHS (EVALUATION)

The chosen independent variables were set in order to gain a result that encompassed a large range of pH, which extended from strongly acidic (pH 0.99) to strongly basic (pH 11.65). This broad range allowed the investigation to observe how caffeine extraction behaved across almost the entire pH scale, making it easier to identify overall trends in the relationship between pH and caffeine mass. Furthermore, the control variables that were set in the beginning of the exploration were properly followed in an attempt to decrease the amount of unknown variables that may take part in the experiment, such as the constant volume, brand, and degassing procedure of the soda. However, there are some errors that could be identified:

LIMITATIONS AND IMPROVEMENTS

Error Category	Description	Improvements
Systematic Error	The large size of the percentage error of the caffeine, which was -30%, may be attributed to the unmeasurable error of Steps 9-11, where the crystalized caffeine was transferred with ethyl acetate to determine the mass of the caffeine. This transfer did not allow for all the caffeine to be weighed in the final beaker, and thus a large portion of this error is hypothesized to have come from this.	To reduce this loss, the beaker used for evaporation should be used throughout the entire process. This would eliminate the need to transfer the caffeine between containers and reduce the loss of solid caffeine during handling.
Random Error	A reliable experimental analysis typically requires at least three trials for each value of the independent variable. But, only one trial was performed for each pH value due to the long evaporation time required for ethyl acetate.	The investigation could be improved by performing at least three trials for each pH value and calculating an average caffeine mass. This would reduce the influence of random errors.
Systematic Error	The scale used to measure the beaker masses had an uncertainty of ± 0.001 g, which is relatively large compared to the very small masses of caffeine extracted. This resulted in very large percentage uncertainties for some data points.	Using an analytical balance with greater precision would significantly reduce the percentage uncertainty in the caffeine mass measurements and improve the accuracy of the results. If the scale were to have even just one more decimal place, this uncertainty would decrease by a factor of ten.

EXTENSION

The investigation could be extended by examining how other factors influence caffeine extraction, such as the volume of ethyl acetate used or the number of extraction steps performed. Experimentation using these variables could provide insight into how solvent interactions and extraction conditions affect the amount of caffeine within carbonated beverages. Another extension would involve measuring caffeine concentration using UV-Visible spectroscopy.

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